msds-422-final-project

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1 MSDS 422-0 – Final Project

1.1 Milestone 2 Submission

1.1.1 Predicting YouTube Video Virality Using Content, Creator, and Engagement Attributes

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Course: MSDS 422 – Machine Learning
Framework: CRISP-DM / TDSP
Dataset: YouTube Videos Data for ML and Trend Analysis (Kaggle)
https://www.kaggle.com/datasets/cyberevil545/youtube-videos-data-for-ml-and-trend-analysis

1.2 1. Executive Summary

1.3 2. Problem Statement and Research Objectives

1.3.1 Research Objectives

- Identify which content attributes are most strongly associated with YouTube video virality
- Examine the relative importance of creator-level, visual, textual, and sentiment-based features
- Assess which predictors are informative prior to widespread audience engagement
- Provide actionable insights for content creators and digital marketing stakeholders

1.4 3. Business Context and Unit of Analysis

YouTube is a dominant digital media platform where visibility and engagement directly influence creator revenue, advertising value, and information diffusion. Understanding what drives video virality is therefore of significant business relevance to content creators, platform strategists, and digital marketers.

The unit of analysis in this project is the individual YouTube video. Viral success is operationalized using engagement-related outcomes such as views, likes, and comments, as observed across trending and non-trending content.

1.5 4. Literature Review

Prior research on social media virality highlights the multifaceted nature of content diffusion, emphasizing the interaction between creator-level attributes, content presentation, and audience engagement. Studies grounded in social capital theory demonstrate that creator characteristics such as subscriber count and content history significantly influence engagement outcomes. Other work emphasizes the role of visual attributes, including image memorability and controlled atypicality, in driving attention and interaction, particularly when aligned with platform and category norms.

Recent machine learning–based studies further show that textual metadata and sentiment signals extracted from titles, descriptions, and comments provide additional predictive power, especially when combined with engagement metrics. Collectively, this literature motivates a multi-view analytical approach that integrates visual, textual, social, and engagement-based features to model YouTube video virality. The present study builds on these findings by empirically evaluating the relative importance of these feature groups within a unified machine learning framework.

1.5.1 4.1 Visual Attributes and Content Presentation

1.5.2 4.2 Creator Social Capital and Engagement Dynamics

1.5.3 4.3 Textual Metadata and Multi-View Learning

1.6 5. Analytical Framework: CRISP-DM

This project follows the CRISP-DM framework to structure the end-to-end machine learning pipeline:

1. Business Understanding – Defining the virality prediction problem and stakeholder objectives
2. Data Understanding – Exploring YouTube video metadata and engagement patterns
3. Data Preparation – Feature engineering, transformation, and scaling
4. Modeling – Training and evaluating multiple ML/DL models
5. Evaluation – Comparing model performance and validating assumptions
6. Deployment – Proposing an automated prediction pipeline

1.7 6. Data Understanding and Exploratory Data Analysis (EDA)

This section explores the structure and characteristics of the YouTube dataset, including feature distributions, missing values, correlations, and initial engagement patterns. Visualizations are used to highlight trends in views, likes, comments, and creator-level attributes.

1.7.1 6.1 Data Loading

1.7.2 6.2 Initial Exploration

1.7.3 6.3 Missingness and Distribution

1.8 7. Data Preparation and Feature Engineering

Feature engineering focuses on transforming raw YouTube metadata into analytically meaningful predictors. Features are grouped into creator-level, visual, textual, and engagement-based categories. Transformations include scaling, normalization, and encoding where appropriate.

1.9 8. Modeling Strategy

The final project will evaluate at least four machine learning models, including linear, tree-based, and non-linear approaches. Model selection will emphasize interpretability, performance, and business relevance.

1.10 9. Milestone 2 Deliverables

This milestone includes the finalized problem statement, executive summary, annotated literature review, exploratory data analysis, and feature engineering plan. All artifacts are maintained in a shared GitHub repository to support collaboration and reproducibility.

1.11 10. References